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# Information Theory and thermodynamic properties of diatomic molecules using Molecular potential

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**Information Theory and thermodynamic properties of diatomic molecules using Molecular potential**

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**Abstract**

In this paper, the bound state solution of the non-relativistic wave equation with a molecular potential function has been obtained in a closed-form using the Nikiforov-Uvarov method. The solutions of the bound state are then applied to study the information-theoretic measures such as the one-dimensional Shannon and Renyi entropic densities. The expectation values for $\langle r \rangle$, $\langle r^2 \rangle$, and $\langle p^2 \rangle$ were obtained to verify Heisenberg's uncertainty principle. Utilizing the energy spectrum equation, the thermodynamic vibrational partition function is obtained via the Poisson summation. Other thermodynamic function variations with absolute temperature have been obtained numerically for four diatomic molecules ($H_2$, $N_2$, $O_2$, and HF). The Shannon global entropic sum inequality has also been verified. The Renyi sum for constrained index parameters satisfies the global entropic inequality. The thermodynamic properties of the four molecules are similar and conform to works reported in the existing literature. The obtained vibrational energies are in fair agreement with the ones obtained using other forms of potential energy. The result further indicates that the lowest bounds for the Shannon, Renyi, and Heisenberg inequalities are ground states phenomena.

## 1.0 Introduction

In recent times, the phase space representation of quantum mechanical systems has enjoyed renewed interest in the measurement of global information or theoretic quantities. The Heisenberg uncertainty principle [1] forms the basic properties of these phase-space representations in quantum mechanics. By using this principle, several researchers have computed the theoretic quantities for many interesting properties for different potential models [2-12]. In 1925 the concept of fisher information was introduced using the Heisenberg relation [13]. The Fisher information which is a measure of the local fluctuation of the variable proved to be a better indicative of the departure of the probability density than the Shannon entropy [14] which measures global information. For a comprehensive analysis of the two theoretic quantities, readers should refer to articles by Dehesa et al. [11] and Gonzalez-Ferez and Dehesa [12]. These quantities have been applied to the complexity of the hydrogenic abstraction reaction in atomic and molecular physics [15, 16]. In solid-state physics, the concept has been used in the calculation of statistical entropic measures in a model of solids [17]. It has also been used in the determination of complex ecological models and ecosystem behaviour in life sciences [18, 19]. As a result of the diverse applications of the theoretic information, it becomes necessary to implement the theoretic information with well-known molecular potentials that the eigensolutions agree with the Rydberg-Klein-Rees (RKR) and spectroscopic experimental data [20-30]. In a related development, the theoretic quantities of some molecular potential have also been studied [3, 31-35]. For instance, the eigensolutions, Shannon Entropy, and Fisher information under the Eckart-Manning-Rosen potential model have been investigated [33]. In ref. [34], the effect of dissociation energy on the Shannon and Renyi entropies was investigated also using the generalized Morse potential model. The eigensolutions techniques, expectation values, and Fisher information of the Wei potential function for the non-relativistic wave equation have also been obtained [35].

On the other hand, the concept of statistical quantum mechanics over the years has assisted in describing the dynamics of particles at the microscopic (ensemble of states) levels. For instance, the improved Manning-Rosen potential was used to predict correctly the Gibbs free energy and enthalpy for some diatomic and tri-atomic gaseous molecules [20-

28]. The simulated results obtained in their studies are in excellent agreement with the experimental data from the national institute of Standards and Technology (NIST) database. Also, the energy spectra and partition function have been obtained using the improved Rosen-Morse and Tietz oscillators respectively to compute the vibrational mean energy, vibrational specific heat, vibrational free energy, and vibrational entropy for diatomic molecular systems [29, 30, 36-41]. Recently a more realistic potential model for diatomic molecules was introduced by Yanar *et al.* [42]. The factorization method was used in obtaining the eigenfunctions and rotational-vibrational energy eigenvalue that agrees perfectly well with the RKR experimental data. Also within the framework of the non-relativistic quantum mechanics, the asymptotic iteration method has been used to obtain the eigensolutions for the general molecular potential [36]. With appropriate an choice of potential parameters, the general molecular potential reduces to Rosen–Morse and Trigonometric Poschl–Teller potentials which have great application for the vibrations of some polyatomic molecules [42]. Owing to the diverse applications of molecular potentials in molecular physics, in this work, we will apply the parametric Nikiforov Uvarov (NU) to obtain the eigensolutions. Also, the position and momentum expectations as well as the Renyi and Shannon information entropies for the $H_2$, $HF$, $N_2$, and $O_2^+$ diatomic molecules will be calculated. The energy level would be used to calculate the canonical partition function and associated thermodynamic properties. This to the best of our knowledge is lacking in literature.
We proposed the molecular potential given by

$$V(r) = D_e\left(\frac{1-e^{2\alpha(r-r_e)}}{1-e^{2\alpha r}}\right)^2 \quad (1)$$

The potential in (1) can be derived by adjusting the potential parameters in the literature [42**,** 43]. The following parameters $D_e$, $r_e$ and $\alpha$ denote the equilibrium dissociation energy, equilibrium bond length and the potential range. The molecular potential obeys the three necessary conditions [44]:

$$\left.\frac{dV(r)}{dr}\right|_{r=r_e} = 0 \quad (2)$$

$$\left.\frac{d^2V(r)}{dr^2}\right|_{r=r_e} = K_e = (2\pi c)^2\mu\omega_e^2 \quad (3)$$

$$V(\infty) - V(r_e) = D_e \quad (4)$$

where $c$, $K_e$, $\omega_e$ and $\mu$ are the respective speed of light, equilibrium force constant, equilibrium vibrational constant and the reduced mass of the molecule. The force constant is obtained as

$$K_e = 8\alpha^2 D_e\left(\frac{e^{2\alpha r_e}}{1-e^{2\alpha r_e}}\right)^2 \quad (5)$$

By combining (3) and (5), the potential range $\alpha$ is determined as

$$\alpha = \frac{1}{2r_e}\left[\pi\omega_e r_e c\sqrt{\frac{2\mu}{D_e}} + W\left(-\pi\omega_e r_e c\sqrt{\frac{2\mu}{D_e}}\,e^{-\pi\omega_e r_e c\sqrt{\frac{2\mu}{D_e}}}\right)\right] \quad (6)$$

where $W(\cdot)$ is the Lambert function or the product logarithm.

**2.0 Bound state solutions of the radial Schrödinger equation using the parametric NU method.**

The solution of the Schrödinger equation enables one to obtain the energy spectra and wave function of soluble and non-soluble potentials. The equation plays a vital role in many aspects of physics since the derived bound states solutions can be applied to describe the physical properties of quantum mechanical systems. The radial SE is given by the equation

$$\frac{-\hbar^2}{2\mu}\frac{dR_{nl}(r)}{dr} + V_{eff}(r)R(r) = E_{nl}R_{nl}(r) \quad (7)$$

where $\hbar(h/2\pi)$ is the reduced Planck's constant. $V_{eff}(r), E_{nl}$ and $R_{nl}(r)$ are the respective effective potential, rotational-vibrational (ro-vibrational) energy spectra and the wave function. The effective potential is given as

$$V_{eff}(r) = V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \quad (8)$$

The analytical solution of (7) with the potential in (1) is not possible except that we make use of an appropriate substitution for the centrifugal term ($1/r^2$). The approximation for $1/r^2$ has been previously used for this potential type in [43]

$$\frac{1}{r^2} = 4\alpha^2\left(c_0 + \frac{e^{-2\alpha r}}{1-e^{-2\alpha r}} + \left(\frac{e^{-2\alpha r}}{1-e^{-2\alpha r}}\right)^2\right), \qquad c_0 = \frac{1}{12} \tag{9}$$

The substitution of the potential in (1) into the SE with a change of variable $s = e^{-2\alpha r}$, results into the Hypergeometric type second-order differential equation.

$$\frac{d^2R_{nl}(s)}{ds^2} + \frac{1-s}{s(1-s)}\frac{dR_{nl}(s)}{ds} + \frac{-\xi_1 s^2 + \xi_2 s - \xi_3}{s^2(1-s)^2}R_{nl}(s) = 0 \tag{10}$$

where

$$\xi_1 = \frac{\mu}{2\alpha^2\hbar^2}(4\alpha^2 c_0 L + D_e e^{4\alpha r_e} - E_{nl}) \tag{11}$$

$$\xi_2 = \frac{\mu}{2\alpha^2\hbar^2}(8\alpha^2 c_0 L - 4L\alpha^2 + 2D_e e^{2\alpha r_e} - 2E_{nl}) \tag{12}$$

$$\xi_3 = \frac{\mu}{2\alpha^2\hbar^2}(4\alpha^2 c_0 L + D_e - E_{nl}) \tag{13}$$

The notation $L$ in eqs. (11-13) is defined as $L = \frac{l(l+1)\hbar^2}{2\mu}$

Equation (10) allows us to obtain the bound state solution using the parametric Nikiforov-Uvarov (PNU) approach (see detailed discussion in [45]). Using the parameters of the PNU approach, the necessary conditions for the respective energy equation and wave function are obtained as

$$\xi_3 = \frac{1}{4}\left(n + \frac{1}{2} + \sqrt{\xi_3 + \xi_1 - \xi_2 + \frac{1}{4}} + \frac{\xi_3 - \xi_1}{n + \frac{1}{2} + \sqrt{\xi_3 + \xi_1 - \xi_2 + \frac{1}{4}}}\right)^2 \tag{14}$$

$$R_{nl}(s) = N_{nl}s^{\sqrt{\xi_3}}(1-s)^{\sqrt{\xi_3+\xi_1-\xi_2+\frac{1}{4}}}P_n^{\left(2\sqrt{\xi_3},2\sqrt{\xi_3+\xi_1-\xi_2+\frac{1}{4}}\right)}(1-2s) \tag{15}$$

where $N_{nl}$ and $P_n^{\left(2\sqrt{\xi_3},2\sqrt{\xi_3+\xi_1-\xi_2+\frac{1}{4}}\right)}(1-2s)$ are the normalization constant and the Jacobi polynomial of order $n$. Inserting the terms $\xi_i(i = 1,2,3)$ in (11)-(13), the energy equation and wave function are obtained in closed form

$$E_{nl} = \Lambda_0 - \Lambda_1\left(n + \Delta + \frac{\Lambda_2}{n+\Delta}\right)^2 \tag{16}$$

$$R_{nl}(r) = N_{nl}(e^{2\alpha r})^{\varrho_1}(1-e^{2\alpha r})^{\varrho_2+1/2}P_n^{(2\varrho_1,2\varrho_2)}(1-2e^{2\alpha r}) \tag{17}$$

Where

$$\Lambda_0 = D_e + \frac{2l(l+1)c_0\alpha^2\hbar^2}{\mu}$$

$$\Lambda_1 = \frac{\alpha^2\hbar^2}{2\mu}$$

$$\Lambda_2 = \frac{\mu D_e\left(1-e^{4\alpha r_e}\right)}{2\alpha^2\hbar^2}$$

$$\Delta = \frac{1}{2} + \sqrt{\left(l+\frac{1}{2}\right)^2 + \frac{\mu D_e(1-e^{2\alpha r_e})^2}{2\alpha^2\hbar^2}}$$

$$\varrho_1 = \sqrt{\frac{\mu}{2\alpha^2\hbar^2}(4\alpha^2 c_0 L + D_e - E_{nl})}$$

$$\varrho_2 = \Delta - \frac{1}{2}$$

The normalization factor $(N_{nl})$ can be obtained from the normalization condition of the radial wave function in the region of space $(0 < r < \infty)$.

$$\int_0^\infty |R_{nl}(r)|^2 dr = 1 \tag{18}$$

To evaluate the integral in (18), we can either use the analytical treatise in [46, 47] or simply use Mathematica programme. For $n = 0, n = 1$, the normalized radial wave functions are obtained as

$$R_{0l}(r) = \sqrt{\frac{2\alpha\Gamma(2\varrho_1+2\varrho_2+2)}{\Gamma(2\varrho_1)\Gamma(2\varrho_2+2)}}(e^{2\alpha r})^{\varrho_1}(1-e^{2\alpha r})^{\varrho_2+1/2} \tag{19}$$

$$R_{1l}(r)=\sqrt{\frac{2\alpha\Gamma(2\varrho_1+2\varrho_2+3)}{(2\varrho_1+1)\Gamma(2\varrho_1)\Gamma(2\varrho_2+3)}}(e^{2\alpha r})^{\varrho_1}(1-e^{2\alpha r})^{\varrho_2+1/2}P_1^{(2\varrho_1,2\varrho_2)}(1-2e^{2\alpha r}) \quad (20)$$

The momentum space wave function can be obtained from the Fourier transform

$$R_{nl}(p)=\frac{1}{\sqrt{2\pi}}\int_0^\infty R_{nl}(r)e^{-ipr}dr \quad (21)$$

For $n=0, n=1$, the normalized momentum space wave function is obtained as

$$R_{0l}(p)=\sqrt{\frac{2\alpha\Gamma(2\varrho_1+2\varrho_2+2)}{2\pi\Gamma(2\varrho_1)\Gamma(2\varrho_2+2)}}\times\frac{\Gamma\left(\varrho_2+\frac{3}{2}\right)\Gamma\left(\varrho_1+\frac{ip}{2\alpha}\right)}{2\alpha\Gamma\left(\varrho_1+\varrho_2+\frac{3}{2}+\frac{ip}{2\alpha}\right)} \quad (22)$$

$$R_{1l}(p)=\sqrt{\frac{2\alpha\Gamma(2\varrho_1+2\varrho_2+3)}{2\pi(2\varrho_1+1)\Gamma(2\varrho_1)\Gamma(2\varrho_2+3)}}\times\frac{\left[2\alpha\left(\varrho_1+\varrho_2+\frac{3}{2}\right)-(2\varrho_2+2)ip\right]\Gamma\left(\varrho_2+\frac{3}{2}\right)\Gamma\left(\varrho_1+\frac{ip}{2\alpha}\right)}{4\alpha^2\Gamma\left(\varrho_1+\varrho_2+\frac{5}{2}+\frac{ip}{2\alpha}\right)} \quad (23)$$

**3. 0 Applications of the bound state solutions**

The analytical expressions for the eigenvalue and the normalized eigenfunction obtained using the new potential model will be tested statistically to verify the correctness of our result. The uncertainty relations as well as the theoretic measures of some diatomic molecules will be undertaken. Also, applications to thermodynamic properties will be done in the preceding section.

**3.1 Uncertainty relations and information theoretic measures.**

Using the normalized wave function, the Heisenberg uncertainty principle can be verified from the knowledge of the radial and momentum space expectation values. The uncertainty plays a vital role in quantum mechanics and tells us that the precise measurement of the momentum of a particle leads to uncertainty in the measurement of its position or vice-visa. The basic ideal of the principle is given by the fundamental relation

$$\Delta r\times\Delta p\geq\frac{\hbar}{2} \quad (24)$$

The position expected values ($\langle r^k\rangle$) is obtained from the integral:

$$\langle r^k\rangle=\int_0^\infty r^k|R_{nl}(r)|^2dr \quad (25)$$

Where $k$ is an integer.

The momentum expected values ($\langle p^k\rangle$) is obtained from the integral

$$\langle p^k\rangle=\int_{-\infty}^\infty p^kR_{nl}^*(p)R_{nl}(p)dp \quad (26)$$

The uncertainties in position and momentum are given by

$$\Delta r=\sqrt{\langle r^2\rangle-\langle r\rangle^2} \quad (27)$$

$$\Delta p=\sqrt{\langle p^2\rangle-\langle p\rangle^2} \quad (28)$$

Also, the normalized radial and momentum space densities can be used to obtain the information theoretic quantities. The Renyi entropies' in position and momentum representations are given as

$$R_q(\rho_{nl})=\frac{1}{1-q}ln\left(\int_0^\infty[\rho_{nl}(r)]^qdr\right) \quad (29)$$

$$R_q(\gamma_{nl})=\frac{1}{1-q}ln\left(\int_{-\infty}^\infty[\gamma_{nl}(p)]^qdp\right) \quad (30)$$

Where $q\in[0,\infty]$ is a real number. When $q\succ1$, the Rényi entropy decreases indicating that the system is stable and we have more information about the system. For $q\prec1$, the entropy increases indicating the decrease of information [48, 49]. Also, the sum of the Rényi entropies obey the inequality [48]

$$R(T)=R_q(\gamma_{nl})+R_k(\rho_{nl})\geq-\frac{D}{2}\left(\frac{1}{1-q}ln\frac{q}{\pi}+\frac{1}{1-k}ln\frac{k}{\pi}\right) \quad (31)$$

where $D$ is the spatial dimension number and the parameters $q$ and $k$ satisfies the linear constraint [48]

$$\frac{1}{q}+\frac{1}{k}=2\,, \qquad 0.5 \le k < 1 \qquad (32)$$

When the index parameter $q = 1$, the Renyi entropies reduce to the Shannon entropies [14] via the L' Hopital's rule.

$$S(\rho_{nl}) = -\int_0^\infty \rho_{nl}(r) ln\rho_{nl}(r) dr \qquad (33)$$

$$S(\gamma_{nl}) = -\int_{-\infty}^\infty \gamma_{nl}(p) ln\gamma_{nl}(p) dp \qquad (34)$$

The inequality $S(T) = S(\rho_{nl}) + S(\gamma_{nl}) \geq D(1 + ln\pi)$ have been used to verify the Shannon entropies sum for different states of quantum mechanical systems [50, 51]. The position and momentum densities are denoted by $\rho_{nl} = |R_{nl}(r)|^2, \gamma_{nl} = R^*_{nl}(p)R_{nl}(p)$,
where $R^*_{nl}(p)$ represents the complex conjugate.

### 3.2 Thermodynamic properties

The vibrational partition function is required to obtaining the thermodynamic equations and to analyze the thermal properties of gaseous substances, it can be written [52]

$$Z_{\text{vib}} = \sum_{n=0}^{n_{\max}} \mathrm{e}^{-\beta E_n}\,, \qquad (35)$$

where $E_n$ is the pure vibrational state energy ($\ell = 0$), $\beta = (\mathrm{k_B}T)^{-1}$, $\mathrm{k_B}$ is the Boltzmann constant, $T$ is the temperature of the diatomic molecules, $n_{\max}$ is the upper bound vibrational quantum number. $n_{\max}$ is obtained by maximizing the energy of the system, that is letting $E'_n(n_{\max}) = 0$. Inserting the energy equation into this expression leads to

$$n_{\max} = \sqrt{\Lambda_2} - \Delta\,. \qquad (36)$$

Substituting the pure vibrational state energy into (35) gives

$$Z_{\text{vib}} = \mathrm{e}^{-\beta\Lambda_0} \sum_{n=0}^{n_{\max}} f(n)\,, \qquad (37)$$

where

$$f(n) = \mathrm{e}^{\lambda^2\left(n+\Delta+\frac{\Lambda_2}{n+\Delta}\right)^2}\,, \qquad \lambda = (\beta\Lambda_1)^{1/2} \qquad (38)$$

To evaluate the summation in (37), the modified Poisson summation formula can be used [53]. This approach is considered here because of its simplicity and the highly accurate results it produced for other potential models [29, 54, 55]. Some alternative summation equations can be found elsewhere [56, 57]. Given a finite summation with an upper bound $n_{\max}$, the modified Poisson summation can be expressed as [53]

$$\sum_{n=0}^{n_{\max}} f(n) = \frac{1}{2}\left\{f(0) - f(n_{\max}+1)\right\} + \sum_{j=-\infty}^{j=+\infty} \int_0^{n_{\max}+1} f(x)\mathrm{e}^{-\mathrm{i}2\pi jx}\,\mathrm{d}x\,, \qquad (39)$$

Substituting (38) into the right-hand side of (39), one can write

$$\sum_{n=0}^{n_{\max}} f(n) = \frac{1}{2}\left(\mathrm{e}^{a^2} - \mathrm{e}^{b^2}\right) + \int_0^{n_{\max}+1} \mathrm{e}^{\lambda^2\left(n+\Delta+\frac{\Lambda_2}{n+\Delta}\right)^2}\mathrm{d}x + \sum_{j=-\infty}^{j=-1} \int_0^{n_{\max}+1} \mathrm{e}^{\lambda^2\left(n+\Delta+\frac{\Lambda_2}{n+\Delta}\right)^2 - \mathrm{i}2\pi jx}\,\mathrm{d}x + \sum_{j=1}^{j=+\infty} \int_0^{n_{\max}+1} \mathrm{e}^{\lambda^2\left(n+\Delta+\frac{\Lambda_2}{n+\Delta}\right)^2 - \mathrm{i}2\pi jx}\,\mathrm{d}x\,, \qquad (40)$$

where $a = \lambda\left(\Delta + \frac{\Lambda_2}{\Delta}\right)$, $b = \lambda\left(n_{\max}+1+\Delta+\frac{\Lambda_2}{n_{\max}+1+\Delta}\right)$. The last-two terms in (40) are the terms with $j \neq 0$, these are the quantum correction terms [29, 55]. In the moderate to high temperature range, contributions from these terms are relatively small, these terms can be neglected, thus, equation (40) is reduced to

$$\sum_{n=0}^{n_{\max}} f(n) = \frac{1}{2}\left(\mathrm{e}^{a^2} - \mathrm{e}^{b^2}\right) + \int_0^{n_{\max}+1} \mathrm{e}^{\lambda^2\left(x+\Delta+\frac{\Lambda_2}{x+\Delta}\right)^2}\mathrm{d}x\,. \qquad (41)$$

By adopting the procedures given in Ref. [29, 53, 57] to evaluate the integral in (41) and inserting the resulting expression in (37), the vibrational partition function is obtained as

$$Z_{\text{vib}} = \frac{1}{2}\mathrm{e}^{-\beta\Lambda_0}\left\{\mathrm{e}^{a^2} - \mathrm{e}^{b^2} - \frac{\sqrt{\pi}}{2\lambda}\left(\text{Efri}a - \text{Erfi}b\right)\right\}, \tag{42}$$

where Erfi $x$, denotes the usual error function of parameter $x$.

Some useful thermodynamic functions including entropy (S), mean thermal energy (U), vibrational Helmholtz free energy (F), and specific heat capacity (C) can be evaluated through the following equations

$$U = -\frac{\partial}{\partial\beta} In\big(Z(\beta)\big) \tag{43}$$

$$C = k_B\beta^2 \frac{\partial^2}{\partial^2\beta} In\big(Z(\beta)\big) \tag{44}$$

$$S = k_B In\big(Z(\beta)\big) - k_B\beta \frac{\partial}{\partial\beta} In\big(Z(\beta)\big) \tag{45}$$

$$F = -\frac{1}{\beta} In\big(Z(\beta)\big) \tag{46}$$

**4.0 Discussion of Results**

The parameter used in the computation of the Shannon entropy, Uncertainty relations, and eigenvalues are presented in Table 1, also Eq. (6) have been used for the computation of the screening parameter $\alpha$ for the various states of the diatomic molecules. Table 2 shows the comparison of the vibrational energies of HF, $H_2$, and $N_2$ respectively, the results obtained using the modified potential model proposed in this study are in perfect agreement with the study of Ghanbari and Khordad [41] using the Tietz-Hua potential model. Also, the vibrational energies obtained increases with an increase in quantum number. As we can see from the results presented in Tables 3 and 4 for the ground and first excited state position and momentum densities space Shannon and Renyi entropies respectively for the selected molecules, the entropies increase with an increase in the quantum number and obeys the Renyi's sum for constrained index parameters and satisfies the global entropic inequality uncertainty of Eq. (31). Also, in Tables 5 and 6, the position and momentum space expectation values increases with an increase in quantum number and obeys the Heisenberg's uncertainty relations for the selected molecules. The vibrational partition function as a function of temperature has been presented for the various molecules under consideration (see Figure 1a), the partition function asymptotically increased with an increase in temperature with an overlap between hydrogen fluoride and hydrogen molecules. Departure from the asymptotic behaviour occurs at about 300K. The internal energy increases monotonically with temperature with hydrogen and hydrogen fluoride having the highest
value. The asymptotic decrease of internal energy with temperature is seen with an oxygen molecule (Fig. 1b). Figure 1c shows a monotonically decrease of the free energy with temperature with $H_2$ and HF having the highest value of internal energy. The rate of disorderliness of the molecules with temperature is seen in Figure 1d. Asymptotic behaviour was observed from absolute
temperatures up to 300K for all the molecules with overlap occurring between $H_2$ and HF respectively. The molecules increase with an increase in the entropy. The specific heat capacity increases monotonically with an increase in temperature for all the molecules under consideration.

**5.0 Conclusion**

The non-relativistic wave equation has been solved with a proposed molecular potential using the Nikiforov-Uvarov method. The eigenvalue and eigenfunction have been obtained in a closed form and applied to obtain the global theoretic and thermodynamic quantities for some diatomic molecules of interest. The expectation values for both the position and momentum space obtained obeys Heisenberg's uncertainty principle. Also, Shannon and Renyi's sum for constrained index parameters satisfies the global entropic inequalities and has been verified using the various diatomic molecules under consideration. Furthermore, the explicit classical partition function has been derived and other thermodynamic functions such as entropy ($S$), mean thermal energy ($U$), mean free energy ($F$), and specific heat capacity ($C$) were all plotted as a function of temperature for the various molecules considered. The dependence of these functions on temperature has been discussed in detail.

**Competing interest statement**

The authors declared that there are no competing interests regarding the publication of the article.

**Author contribution**
MCO and ANI wrote the original draft. EO and CAO carried out data curation and computations. EO, ESE and IBO validated the results. DAO and OPO carried out discussion of results.

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**Table 1:** Spectroscopic parameters for selected diatomic molecules taken from literature [39, 40]

| Molecule | $D_e(cm^{-1})$ | $r_e(\text{Å})$ | $\mu(g)\times 10^{-23}$ | $\omega_e(cm^{-1})$ | $\alpha(\text{Å}^{-1})$ |
|---|---|---|---|---|---|
| $H_2$ | 38318 | 0.741 | 0.084 | 4407.682 | 0.53163 |
| HF | 49382 | 0.917 | 0.160 | 4138.997 | 0.89927 |
| $N_2$ | 79885 | 1.097 | 1.171 | 2358.587 | 1.26526 |
| $O_2^+$ | 54688 | 1.116 | 1.337 | 1905.175 | 1.33624 |

Table 2: Comparison of the vibrational energies $(E_{n,0}-D)\ (eV)$ for HF, $H_2$, and $N_2$ molecules.

| State | HF | | $H_2$ | | $N_2$ | |
|---|---|---|---|---|---|---|
| $n$ | Present | Ref. [41] | Present | Ref.[41] | Present | Ref. [41] |
| 0 | -5.868631409 | -5.868757846 | -4.482180900 | -4.481571826 | -9.758550309 | -9.7588029855 |
| 5 | -3.680036270 | -3.660498629 | −2.411375939 | -2.281533873 | -8.365087277 | -8.359551147 |
| 7 | -2.968513788 | -2.936429314 | -1.825861680 | -1.629999641 | -7.839502288 | -7.829307157 |

Table 3: Shannon and Renyi information entropies for ground state

| Molecule | Shannon entropy | | | Renyi entropy | | |
|---|---|---|---|---|---|---|
| | $S(\rho_{00})$ | $S(\gamma_{00})$ | $S(T)\geq 2.14473$ | $R_2(\gamma_{00})$ | $R_{2/3}(\rho_{00})$ | $R(T)\geq 2.0995$ |
| $H_2$ | -4.04475 | 6.19800 | 2.15326 | 6.03975 | -3.93601 | 2.10374 |
| HF | -3.89968 | 6.04709 | 2.14741 | 6.1568 | −4.05328 | 2.10353 |
| $N_2$ | -5.00103 | 7.14607 | 2.14504 | 6.99247 | - 4.89282 | 2.09966 |
| $O_2^+$ | - 4.71800 | 6.86302 | 2.14503 | 6.70943 | -4.60978 | 2.09965 |

Table 4: Shannon and Renyi information entropies for first excited state

| Molecule | Shannon entropy | | | Renyi entropy | | |
|---|---|---|---|---|---|---|
| | $S(\rho_{10})$ | $S(\gamma_{10})$ | $S(T)\geq 2.14473$ | $R_2(\gamma_{10})$ | $R_{2/3}(\rho_{10})$ | $R(T)\geq 2.0995$ |
| $H_2$ | -3.71512 | 6.37702 | 2.66190 | 6.23398 | - 3.53884 | 2.69514 |
| HF | -3.56424 | 6.24533 | 2.68109 | 6.10666 | -3.4481 | 2.65856 |
| $N_2$ | -4.70556 | 7.39023 | 2.68467 | 7.25379 | - 4.60965 | 2.64414 |
| $O_2^+$ | - 4.41838 | 7.10340 | 2.68502 | 6.96695 | - 4.32293 | 2.64402 |

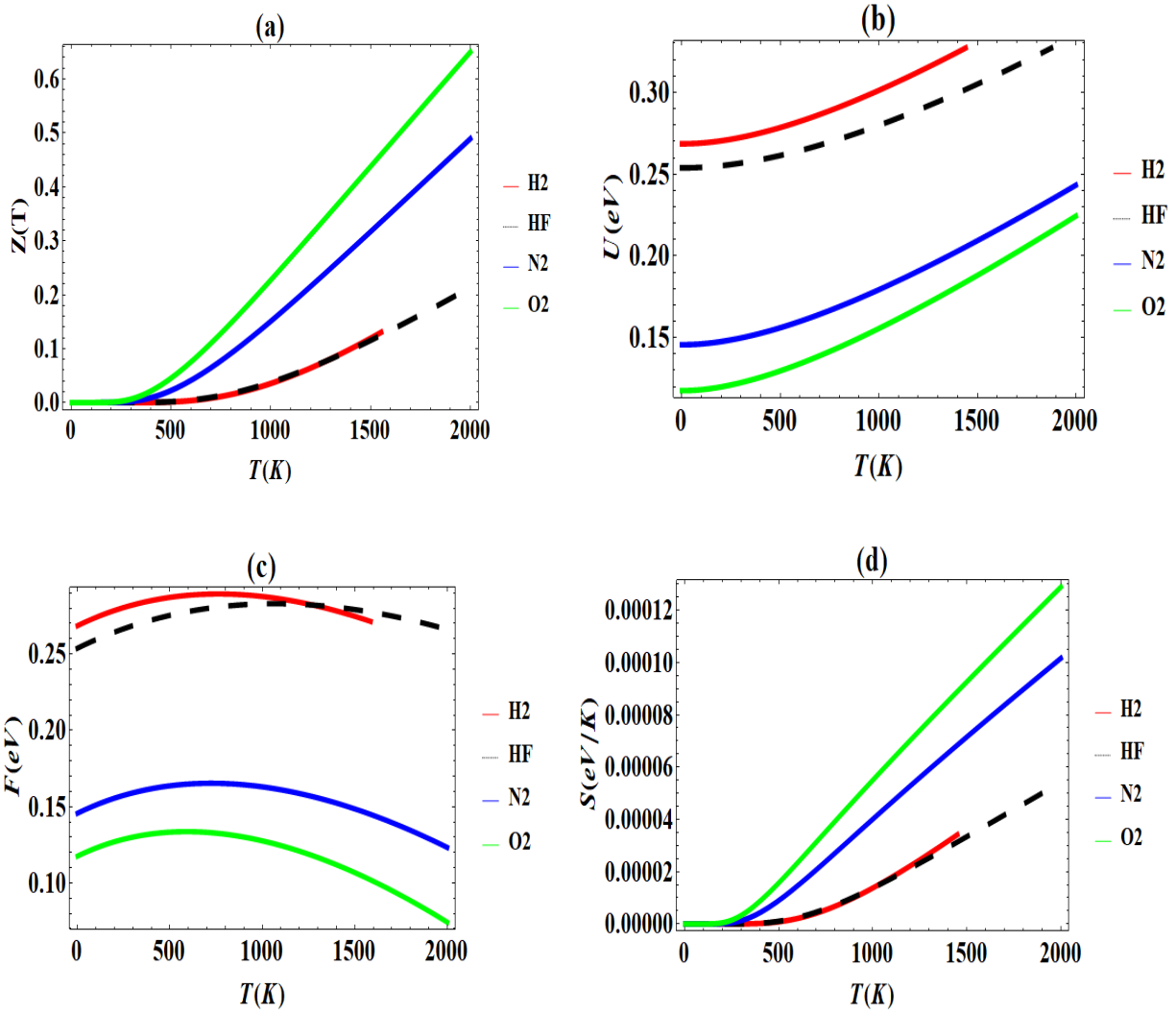
Table 5: Expectation values and Heisenberg's uncertainty for ground state.

| Molecule | Ground state $(n=0, l=0)$ | | | | | | |
|---|---|---|---|---|---|---|---|
| | $\langle r^2\rangle$ | $\langle r\rangle$ | $\Delta r$ | $\langle p^2\rangle$ | $\langle p\rangle$ | $\Delta p$ | $\Delta r\Delta p\geq 0.5$ |
| $H_2$ | 0.00143919 | 0.0376972 | 0.00425602 | 14162.8 | 0 | 119.008 | 0.506499 |
| HF | 0.00603922 | 0.0775574 | 0.00490607 | 10470.6 | 0 | 102.326 | 0.502019 |
| $N_2$ | 0.00576161 | 0.0758878 | 0.00162895 | 94303.0 | 0 | 307.088 | 0.500232 |
| $O_2^+$ | 0.01053910 | 0.1026370 | 0.00216186 | 53539.4 | 0 | 231.386 | 0.500225 |

Table 6: Expectation values and Heisenberg's uncertainty for first excited state.

| Molecule | Excited state $(n=1, l=0)$ | | | | | | |
|---|---|---|---|---|---|---|---|
| | $\langle r^2\rangle$ | $\langle r\rangle$ | $\Delta r$ | $\langle p^2\rangle$ | $\langle p\rangle$ | $\Delta p$ | $\Delta r\Delta p\geq 0.5$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $H_2$ | 0.00206204 | 0.0449382 | 0.00652662 | 33492.6 | 0 | 183.010 | 1.19444 |
| HF | 0.00739649 | 0.0856588 | 0.00768233 | 26672.5 | 0 | 163.317 | 1.25517 |
| $N_2$ | 0.00611308 | 0.0781472 | 0.00246912 | 267828 | 0 | 517.521 | 1.27782 |
| $O_2^+$ | 0.01126230 | 0.1060710 | 0.00334190 | 150904 | 0 | 388.464 | 1.29821 |



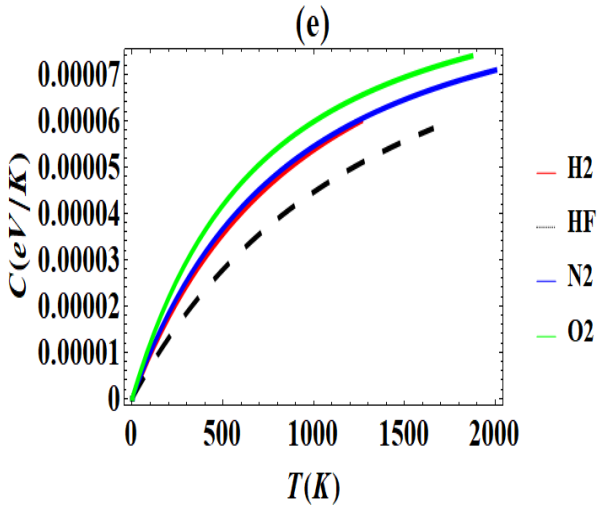



Figure 1. (a)Variation of the vibrational Partition function with temperature. (b) Variation of the Internal energy U with Temperature (c) Variation of the Free energy with temperature (d)Variation of the entropy with temperature (e) Variation of the specific heat capacity with temperature

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